PROJECT PHASE 5: MODEL GENERALIZATION AND SYSTEM PROTOTYPE

Deadline: Week 10
Weight: 10% of the project grade
Penalty: Each delay is penalized with 3 points
Maximum allowed delays: 2

🎯 **Objective**

The goal of this phase is to validate the robustness of the developed solution by applying it to new data and to demonstrate its practical usability through a simple application prototype.

Students must:

- test whether their pipeline generalizes to new datasets
- analyze how performance changes across datasets
- build a minimal application (UI) that exposes the model

This phase focuses on:

- generalization capability (does your model work beyond the original dataset?)
- practical usability (can your solution be used in a real-world scenario?)

This represents the transition from model → system.

💻 Coding (Generalization & Application Prototype)

In this phase, you must extend your work from Phase 4.

1. Apply Pipeline on New Dataset(s)

- Use at least one additional dataset (preferably similar domain)
- You may use:
  - a second dataset from the same source
  - a dataset from a different source but same problem
- The dataset must respect project constraints (medium/large size when possible)

You must:

- adapt the pipeline if necessary
- ensure it runs end-to-end on the new dataset

2. Dataset Adaptation

- Handle differences between datasets:
    - schema differences
    - missing features
    - distribution shifts
- Clearly explain:
    - what needed to be changed
    - what remained reusable

3. Experiments on New Dataset(s)

- Run the best model from Phase 4 on the new dataset
- Optionally compare with:
    - baseline model
    - alternative algorithm

You must include:

- evaluation metrics
- comparison with original dataset results

4. Cross-Dataset Comparison

- Compare performance across:
    - original dataset
    - new dataset(s)

Discuss:

- performance drop or improvement
- possible causes:
    - data distribution differences

- feature mismatch
- noise / quality issues

5. Application Prototype (UI)

You must implement a minimal working application that uses your model.

Accepted formats:

- Web application (Flask / FastAPI / Streamlit)
- Desktop application (simple GUI)

Minimum requirements:

- user input (form / file / parameters)
- model inference (prediction / clustering / recommendation / detection)
- output display (result shown clearly)

Examples:

- upload a transaction → predict fraud
- input customer data → predict churn
- input text → detect toxicity
- select user → get recommendations

⚠️ Important:

- UI does not need to be complex
- focus on functionality, not design
- must demonstrate end-to-end usage

📄 **Report**

- Please continue to work on the report template provided: https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template

1. New Dataset Description

- Description of the new dataset(s):

    - size, structure, features
- Differences compared to original dataset
- Challenges introduced by the new data

2. Pipeline Adaptation

- What changes were required to reuse the pipeline
- What components remained unchanged
- Discussion on pipeline flexibility

3. Experiments on New Dataset(s)

- Evaluation results
- Metrics used (same as Phase 3 & 4 for consistency)
- Tables / plots

4. Cross-Dataset Comparison

- Comparison between:
    - original dataset results
    - new dataset results

Must include:

- interpretation of differences
- discussion of generalization capability

5. Application Prototype

- Description of the application:
    - architecture (simple diagram optional)
    - technologies used
    - use cases

6. Final Insights (Very Important)

- What worked well across datasets
- What failed or degraded
- Lessons learned:
  - about data
  - about models
  - about scalability

📦 Deliverables (Mandatory)

Same as previous phases:

Report:

- **LaTeX source**
- **PDF version**

Code:

- **Jupyter Notebook / Python scripts**
- **Application code (mandatory)**
- **Must be runnable**

Repository:

- **GitHub Classroom (same repo)**

Submission rule:

- **Upload before laboratory in Week 10**
- **Do not push only an archive!**
- **Push the deliverables on the main branch !**

Grading

- Report: 50%
- Implementation (code + application): 50%

Final grade = weighted average

Important

- **Work will NOT be evaluated if not committed to GitHub Classroom**
- **Ensure all required files are present before deadline**

Grading Rubric — Code (Generalization & Application)

Section	Points	Full points	Partial points	No points
1 . Pipeline reuse on new dataset	2	Pipeline correctly adapted and runs end-to-end	Partial adaptation, unclear fixes	Missing
2. Dataset adaptation	1.5	Differences handled correctly and explained in code	Attempted but incomplete	Missing
3. Experiments on new data	2	Correct experiments with meaningful evaluation .	Basic evaluation only	Missing
4 . Cross-dataset comparison	1.5	Clear comparison and interpretation	Results shown but not analyzed	Missing
5. Application prototype	2	Functional UI with model integration	UI exists but limited functionality	Missing
5 . Code clarity	1	Clean, documented	Superficial	Poor
Total	**10 points**			

Grading Rubric — Report Phase 5

Section	Points	Full points	Partial points	No points
1 . New dataset description	2	Clear description + differences explained	Basic description	Missing
2 . Pipeline adaptation	2	Well explained reuse and modifications	Partial explanation	Missing
3 . Experiments & results	2	Clear results with interpretation	Results without insights	Missing
4 . Cross-dataset comparison	2	Strong analysis of generalization	Superficial comparison	Missing
5. Application description	1	Clear explanation of system and usage	Minimal description	Missing
5 . Final insights	1	Insightful conclusions and lessons learned	Basic conclusions	Poor
Total	**10 points**			